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MATHS MASTER _____

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CANDIDATE NUMBER

2024 Trial Examination

Form VI Mathematics Extension 1

Friday 16th August, 2024

12.50 pm

General Instructions

- Reading time — 10 minutes
- Working time — 2 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

Fourteen Questions — 70 Marks

Section I (10 marks) Questions 1 – 10

- This section is multiple-choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (60 marks) Questions 11 – 14

- Each question is worth 15 marks.
- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

Collection

- Your name and master should only be written on this page.
- Write your candidate number on this page, on each booklet and on the multiple choice sheet.
- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- 4 booklets per boy
- Candidature: 126 pupils

Writer: NJL

Section I

Questions in this section are multiple-choice.

Record the single best answer for each question on the provided answer sheet.

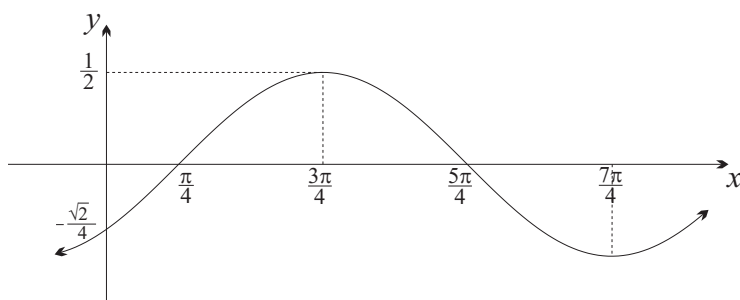
1. Given that there are 2000 employees in a company, which of the following represents the least number of employees that must share the same birthday?

(A) 3
(B) 4
(C) 5
(D) 6

2. What is the amplitude of $y = \sin x - \cos x$?

(A) $-\sqrt{2}$
(B) 1
(C) $\sqrt{2}$
(D) 2

3. Which of the following equations corresponds to the graph shown below?



(A) $y = \frac{1}{2} \sin(x + \frac{\pi}{4})$
(B) $y = \frac{1}{2} \sin(x - \frac{\pi}{4})$
(C) $y = \frac{1}{2} \cos(x - \frac{\pi}{4})$
(D) $y = \frac{1}{2} \cos(x + \frac{\pi}{4})$

4. Given that $\sin \theta = \frac{9}{41}$, and θ is obtuse, which of the fractions shown below is equivalent to $\sin 2\theta$?

(A) $-\frac{720}{1681}$
(B) $-\frac{81}{1681}$
(C) $\frac{81}{1681}$
(D) $\frac{720}{1681}$

5. Which of the following expressions is equal to $\frac{d}{dx} \tan^{-1} \sqrt{1-x}$?

(A) $\frac{1}{2\sqrt{1-2x}}$

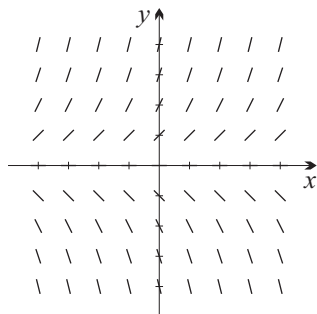
(B) $-\frac{1}{2\sqrt{1-2x}}$

(C) $\frac{1}{2\sqrt{1-x}} \sec^2 \sqrt{1-x}$

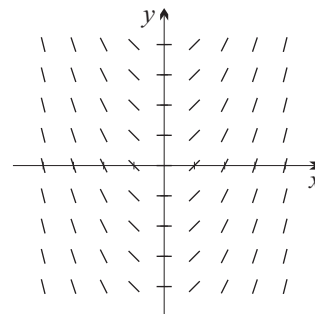
(D) $\frac{1}{2(x-2)\sqrt{1-x}}$

6. Which of the following slope fields corresponds to the differential equation $y' = y - x$?

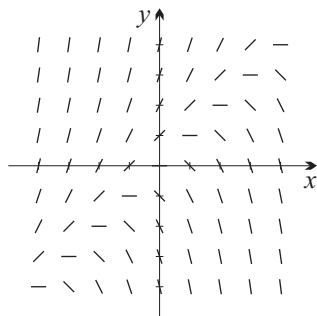
(A)



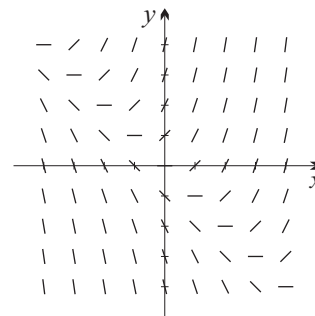
(B)



(C)



(D)



7. Given that $\underline{a} = 2\underline{i} - 3\underline{j}$ and $\underline{b} = -4\underline{i} + \underline{j}$, which of the following expressions represents the unit vector in the direction of $\underline{a} + \underline{b}$?
- (A) $-\frac{1}{\sqrt{2}}(\underline{i} + \underline{j})$
(B) $\frac{1}{\sqrt{5}}(\underline{i} + 2\underline{j})$
(C) $-\frac{1}{\sqrt{5}}(\underline{i} + 2\underline{j})$
(D) $\frac{1}{\sqrt{2}}(\underline{i} + \underline{j})$
8. Which equation below is the general solution to $x + yy' = 0$, where C is an arbitrary constant and $y \neq 0$?
- (A) $y = -\frac{x^2}{2y} + C$
(B) $y = -\frac{x^2}{y^2} + C$
(C) $y = \pm\sqrt{C + x^2}$
(D) $y = \pm\sqrt{C - x^2}$
9. Which of the following expressions is $\int \frac{4}{x}(\ln x)^3 dx$?
- (A) $(\ln x)^4 + C$
(B) $4x^4 \ln x + C$
(C) $\frac{4x^4}{\ln x} + C$
(D) $\frac{4}{x^4} + C$
10. Which differential equation below does not represent a form of the logistic equation?
- (A) $\frac{dN}{dt} = \frac{t}{50} \left(1 - \frac{t}{100}\right)$
(B) $\frac{1}{N} \cdot \frac{dN}{dt} = 12 - \frac{N}{6}$
(C) $\frac{dN}{dt} = \sqrt{3} \left(N - \left(\frac{N}{100}\right)^2\right)$
(D) $\frac{dN}{dt} = \frac{N}{18} \left(2 - \frac{N}{10}\right)$

End of Section I

The paper continues in the next section

Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

QUESTION ELEVEN (15 marks) Start a new answer booklet.

- (a) (i) Find the values of a and b if $x^2 + 4x - 21 = (x + a)^2 - b$. 1
- (ii) Hence find $\int \frac{1}{\sqrt{21 - 4x - x^2}} dx$. 2
- (b) Let $f(x) = (x^2 + k)(2x + 3) + 3$, where k is a constant.
- (i) Write down the remainder when $f(x)$ is divided by $(2x + 3)$. 1
- (ii) Given that the remainder when $f(x)$ is divided by $(x - 2)$ is 24, prove that $k = -1$. 2
- (iii) Hence factorise $f(x)$ completely. 1
- (c) A school play requires at least two boys and exactly twice as many girls as boys. If four boys and six girls audition, how many different casts can be formed? 2
- (d) Find the coefficient of x^4 in the expansion of $(1 + 3x)^4(1 - x)^5$. 3
- (e) Use the substitution $u = x^2 - 3x + 1$ to evaluate $\int_3^5 \frac{2x - 3}{\sqrt{x^2 - 3x + 1}} dx$. Give your answer as an exact value. 3

QUESTION TWELVE (15 marks) Start a new answer booklet.

- (a) Let $f(x) = \cos^{-1} x$. Sketch the graphs of the following, clearly showing intercepts and endpoints. Your graphs should be on separate number planes and be about one-third of a page each.

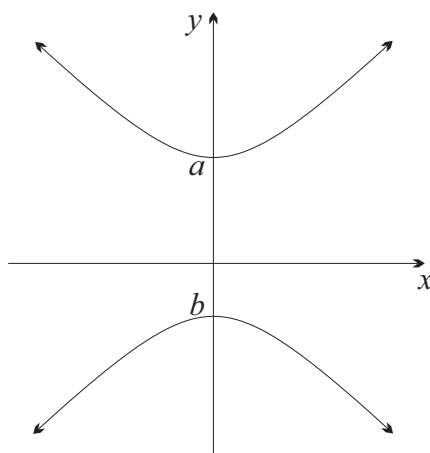
(i) $y = f(x)$

1

(ii) $|y| = f(|x|)$

1

- (b) The graph below has y -intercepts a and b and is defined by the parametric equations $x = 3 \tan \theta$ and $y = 1 - 3 \sec \theta$.



- (i) Find the Cartesian equation of the graph.

2

- (ii) Hence, or otherwise, find the values of a and b .

2

- (c) At time $t = 0$, a football is kicked and leaves the ground with speed 20 m/s at an angle of projection of 30° to the horizontal. Assume that upwards and to the right are positive and that the magnitude of acceleration due to gravity is 10 m/s^2 .

- (i) If the initial velocity is given by the vector $\underline{V} = a\underline{i} + b\underline{j}$, find the values of a and b .

2

- (ii) How high is the ball above the ground when it has travelled $5\sqrt{3}$ metres horizontally?

2

- (iii) What is the exact speed of the ball, and the angle of the ball's velocity to the horizontal, when $t = 1.5 \text{ s}$?

2

- (d) Let $\underline{a} = \lambda\underline{i} + 3\underline{j}$ and $\underline{b} = 6\underline{i} - 2\underline{j}$. The length of the projection of $\underline{a}$ onto $\underline{b}$ is $3\sqrt{10}$. Find the possible values of λ .

3

QUESTION THIRTEEN (15 marks) Start a new answer booklet.

- (a) Find the exact volume of revolution when the region bounded by the curve $y = \cos x$ and the x -axis, between $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$, is rotated about the x -axis. 3

- (b) A spherical water purification tablet is added to a tank of water. After t hours, the tablet is a sphere with radius r mm, surface area A mm² and volume V mm³. The tablet dissolves with a rate of change of volume directly proportional to its surface area, such that

$$\frac{dV}{dt} = -k_1 A,$$

where k_1 is a positive constant. Let V_0 be the volume at $t = 0$.

You may assume the formulae $V = \frac{4}{3}\pi r^3$ and $A = 4\pi r^2$.

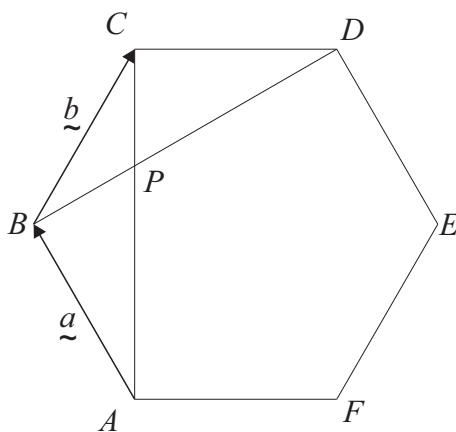
- (i) Show that $\frac{dV}{dt} = -k_2 V^{\frac{2}{3}}$, where $k_2 = k_1(36\pi)^{\frac{1}{3}}$. 2

- (ii) Solve $\frac{dV}{dt} = -k_2 V^{\frac{2}{3}}$ to obtain V in terms of t , k_2 and V_0 . 2

- (iii) In six hours $\frac{7}{8}$ of the volume of the tablet has dissolved. Express k_2 in terms of V_0 . 2

- (iv) How long, to the nearest minute, does it take for 99% of the volume of the tablet to dissolve? 1

- (c) The diagram below shows the regular hexagon $ABCDEF$. Let $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{BC} = \underline{b}$. Interval BD intersects AC at P where $\overrightarrow{AP} = \lambda \overrightarrow{AC}$.



- (i) Express the ratio $|\overrightarrow{AP}| : |\overrightarrow{PC}|$ in terms of λ . 1
- (ii) Find an expression for $\overrightarrow{AP}$ in terms of $\underline{a}$, $\underline{b}$ and λ . 1
- (iii) Use vector methods and the fact that $ABCDEF$ is a regular hexagon, to express $\overrightarrow{BD}$ in terms of $\underline{a}$ and $\underline{b}$. 1
- (iv) Hence use vector methods to find the exact value of λ . 2

QUESTION FOURTEEN (15 marks) Start a new answer booklet.

- (a) A drone flies over a city at a constant altitude of 800 m and velocity of 108 km/h. A soldier on the ground waits until the drone is directly above her before firing a bullet that leaves the gun at a speed of 300 m/s. You may assume that the magnitude of the acceleration due to gravity is 10 m/s^2 .

(i) At what angle to the horizontal must she fire the bullet in order to hit the drone? 2

(ii) How high above the ground must the drone be so that it is too high to shoot down? 3

- (b) The rate of change of a tadpole population is given by the differential equation

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{k} \right)$$

where P is the number of tadpoles after t days, and r and k are positive constants. Initially, the pond is home to P_0 tadpoles.

(i) Show that the tadpole population is increasing when $0 < P < k$. 1

(ii) Solve the differential equation to find P as a function of t . 3

You may assume $\frac{k}{P(k-P)} = \frac{1}{P} + \frac{1}{k-P}$.

(iii) Find $\lim_{t \rightarrow \infty} P$, which is the maximum number of tadpoles that the pond can support. 1

(iv) Find an expression for the time when the rate of change of population is greatest. 2

- (c) Consider the sequence $T_n = \sin(x + (n-1)y)$ for integers $n \geq 1$. 3

Use mathematical induction to show that, for $n \geq 1$,

$$T_1 + T_2 + T_3 + \dots + T_n = \frac{\sin(x + \frac{1}{2}(n-1)y) \sin(\frac{1}{2}ny)}{\sin \frac{1}{2}y}.$$

————— **END OF PAPER** —————

Sydney Grammar School

Extension I Trial Examination 2024

- 1.) 2000 employees, 365 days.
 $5 \times 365 = 1825$
 $6 \times 365 = 2190$
 $\therefore$ least number of employees that share the same birthday is 6.

D

2.) $y = \sin x - \cos x = R \cos(x - \alpha)$

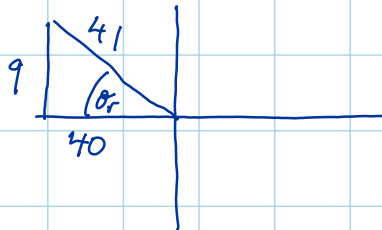
$$R = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$$

C

3.) $y = \frac{1}{2} \sin(x - \frac{\pi}{4})$

B

4.) $\sin \theta = \frac{9}{41}$ $96^\circ < \theta < 180^\circ \therefore$ 2nd quadrant.



$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin \theta = \frac{9}{41} \quad \cos \theta = -\frac{40}{41}$$

$$\sin 2\theta = 2 \times \frac{9}{41} \times -\frac{40}{41}$$

$$= -\frac{720}{1681}$$

A

5.)

$$\frac{d}{dx} \tan^{-1} \sqrt{1-x}$$

$$u = (1-x)^{1/2}$$

$$u' = \frac{1}{2} (1-x)^{-1/2}$$

$$= \frac{1}{2\sqrt{1-x}}$$

$$\frac{d}{dx} \tan^{-1} u = \frac{u'}{1+u^2}$$

$$= \frac{\frac{-1}{2\sqrt{1-x}}}{1 + [\sqrt{1-x}]^2}$$

$$= \frac{\frac{-1}{2\sqrt{1-x}}}{1 + 1 - x}$$

$$= \frac{\frac{-1}{2\sqrt{1-x}}}{2(2-x)\sqrt{1-x}}$$

$$= \frac{1}{2(x-2)\sqrt{1-x}}$$

D

6.) $y' = y - x$

C

7.) $\underline{a} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ $\underline{b} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$ $\underline{a} + \underline{b} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

$$|\underline{a} + \underline{b}| = \sqrt{(-2)^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$$

$$\therefore \frac{1}{2\sqrt{2}} (-2\underline{i} - 2\underline{j})$$

$$= -\frac{2}{2\sqrt{2}} (\underline{i} + \underline{j})$$

$$= -\frac{1}{\sqrt{2}} (\underline{i} + \underline{j})$$

A

$$8.) \quad x + yy' = 0 \quad y \neq 0$$

$$yy' = -x$$

$$y \frac{dy}{dx} = -x$$

$$\int y \, dy = \int -x \, dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + C$$

$$y^2 = -x^2 + C$$

$$y = \pm \sqrt{C - x^2}$$

D

$$9.) \quad \int f'(x) [f(x)]^n \, dx = \frac{1}{n+1} [f(x)]^{n+1} + C$$

$$4 \int \frac{1}{x} [\ln(x)]^3 \, dx = 4 \times \frac{1}{4} [\ln(x)]^4 + C$$

$$= (\ln(x))^4 + C$$

A

$$10.) \quad \frac{dN}{dt} = \frac{t}{50} \left(1 - \frac{t}{100} \right) \quad \text{is not of the form}$$

$$\frac{dN}{dt} = rN(P - N)$$

A

11)a)

$$\begin{aligned} \text{i)} \quad & x^2 + 4x - 21 \\ &= (x+2)^2 - 4 - 21 \\ &= (x+2)^2 - 25 \end{aligned}$$

$$a=2 \quad b=25$$

$$\text{ii)} \quad \int \frac{1}{\sqrt{-(x^2 + 4x - 21)}} dx$$

$$= \int \frac{1}{\sqrt{-(x+2)^2 - 25}} dx$$

$$= \int \frac{1}{\sqrt{25 - (x+2)^2}} dx$$

$$= \sin^{-1} \frac{x+2}{5} + C$$

Working mark for recognising $\sin^{-1}$ of some function.

b) i) $\frac{(x^2+k)(2x+3)+3}{2x+3} = x^2+k \text{ remainder } 3. \checkmark$

ii) $f(2) = 24 \checkmark$
 $24 = (4+k)(4+3) + 3$
 $24 = (4+k) \times 7 + 3$
 $21 = 7(4+k)$
 $3 = 4+k$
 $k = -1$ $\checkmark$

iii) $f(x) = (x^2-1)(2x+3) + 3$
 $= 2x^3 + 3x^2 - 2x - 3 + 3$
 $= 2x^3 + 3x^2 - 2x$
 $= x(2x^2 + 3x - 2)$
 $= x(2x-1)(x+2) \checkmark$

c) At least 2 boys, exactly twice as many girls.

boys	girls
2	4
3	6

for either case identified.

${}^4C_2 \times {}^6C_4 + {}^4C_3 \times {}^6C_6 = 94 \checkmark$
 $\swarrow \quad \searrow$
 either $\checkmark$

$$d.) \quad (1+3x)^4 = {}^4C_0 + {}^4C_1(3x) + {}^4C_2(3x)^2 + {}^4C_3(3x)^3 + {}^4C_4(3x)^4$$

$$(1-x)^5 = {}^5C_0 + {}^5C_1(-x) + {}^5C_2(-x)^2 + {}^5C_3(-x)^3 + {}^5C_4(-x)^4 + {}^5C_5(-x)^5$$

terms in x^4 :

$${}^4C_0 \times {}^5C_4(-x)^4 + {}^4C_1(3x) \times {}^5C_3(-x)^3 + {}^4C_2(3x)^2 \times {}^5C_2(-x)^2$$

$$+ {}^4C_3(3x)^3 \times {}^5C_1(-x) + {}^4C_4(3x)^4 \times {}^5C_0$$

✓ for any one of correct 5 cases seen, above.

$$= 1 \times 5x^4 + 12x(-10x^3) + 54x^2(10x^2) + 108x^3(-5x) + 81x^4 \quad \checkmark$$

$$= 5x^4 - 120x^4 + 540x^4 - 540x^4 + 81x^4$$

$$\text{coefficient} = \underline{\underline{-34}} \quad \checkmark$$

$$e) \quad \int_3^5 \frac{2x-3}{\sqrt{x^2-3x+1}} dx$$

$$u = x^2 - 3x + 1$$

$$\frac{du}{dx} = 2x - 3$$

$$\frac{du}{2x-3} = dx$$

$$5 \rightarrow //$$

$$3 \rightarrow 1$$

$$= \int_1^// \frac{2x-3}{\sqrt{x^2-3x+1}} \frac{du}{(2x-3)}$$

$$= \int_1^// u^{-1/2} du$$

$$= 2 \left[u^{1/2} \right]_1^// \quad \checkmark$$

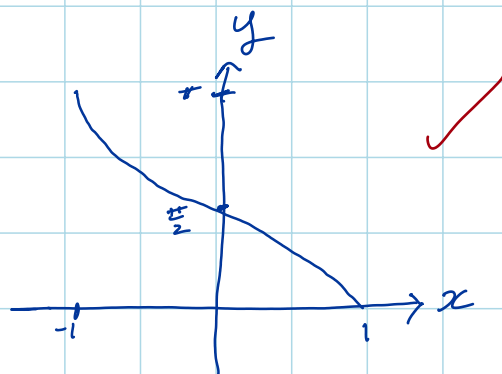
$$= 2(\sqrt{11} - 1) \quad \checkmark$$

(factorisation not req.)

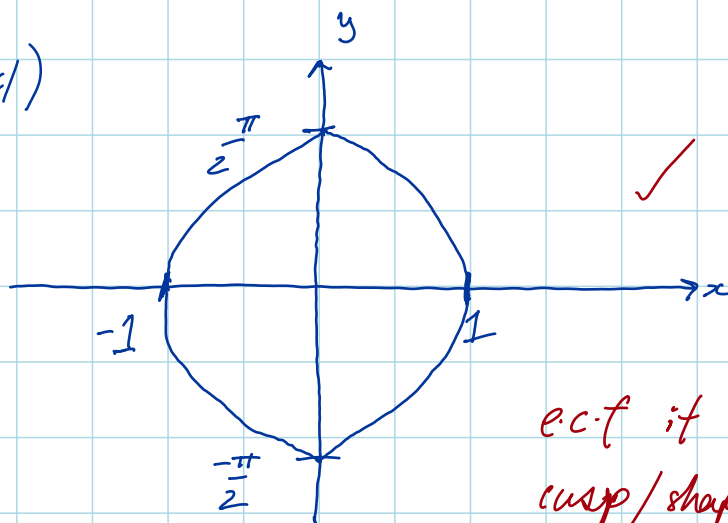
✓ or
equival.
limits req.

12) a) i)

$$y = \cos^{-1}(x)$$



i) $|y| = \cos^{-1}(|x|)$



e.c.f if i) incorrect.
cusp/shape

Don't deduct missing x and/or y axes.
Do deduct for any missing values as per above.

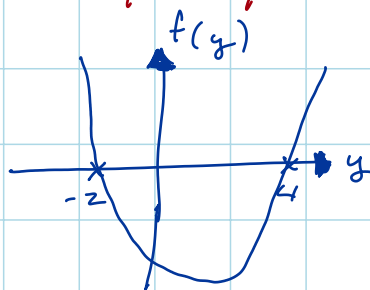
b) i) $x = 3\tan\theta$
 $x^2 = 9\tan^2\theta$ ✓
 $x^2 = 9(\sec^2\theta - 1)$

$y = 1 - 3\sec\theta$
 $3\sec\theta = 1 - y$
 $9\sec^2\theta = (1 - y)^2$
 $9\sec^2\theta - 9 = (1 - y)^2 - 9$
 $9(\sec^2\theta - 1) = (1 - y - 3)(1 - y + 3)$

$x^2 = (-2 - y)(4 - y)$

$x^2 = (y + 2)(y - 4)$ ✓ (or equiv)

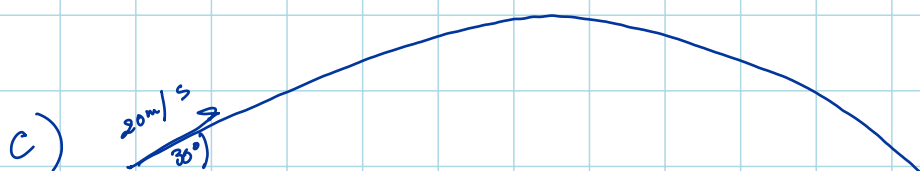
ii) $x = \pm \sqrt{(y + 2)(y - 4)}$
 $(y + 2)(y - 4) \geq 0$



$y \geq 4$ or $y \leq -2$

$a = 4$, $b = -2$ ✓ (Both)

✓ working mark if substitute $x = 0$ into cartesian or parametric equations.



i)

initially; $\underline{V} = 20 \cos 30 \underline{i} + 20 \sin 30 \underline{j}$

$\underline{V} = 10\sqrt{3} \underline{i} + 10 \underline{j}$

$a = 10\sqrt{3} \quad b = 10$

✓

✓

(Both required for 2nd mark).

ii) $\dot{x} = 10\sqrt{3}$

$x = 10\sqrt{3}t$

Let $x = 5\sqrt{3}$

$5\sqrt{3} = 10\sqrt{3}t$

$t = \frac{1}{2} \text{ s.}$

✓

$\ddot{y} = -10$

$\dot{y} = -10t + c$

$t = 0, \dot{y} = 10, \therefore c = 10$

$\dot{y} = -10t + 10$

$y = -5t^2 + 10t + D$

$t = 0, y = 0, \therefore D = 0$

$y = -5t^2 + 10t$

at $t = \frac{1}{2}$

$y = -5\left(\frac{1}{2}\right)^2 + 10\left(\frac{1}{2}\right)$

$y = 3\frac{3}{4} \text{ m}$

✓

(units not required)

$$\text{iii)} \quad t = 1.5 \quad x = 10\sqrt{3}$$

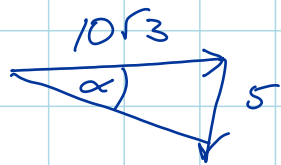
$$y = -10 \times 1.5 + 10$$

$$y = -5$$

$$\begin{aligned} \text{speed} &= \sqrt{(10\sqrt{3})^2 + (-5)^2} \\ &= \sqrt{325} \text{ m/s} \\ &= 5\sqrt{13} \text{ m/s.} \end{aligned}$$



let angle to horizontal be α .



$$|\tan \alpha| = \frac{5}{10\sqrt{3}}$$

$$\alpha = 16.102$$

$$\therefore \text{Angle to horizontal} = -16^\circ$$



(To nearest degree).

(must be negative, any reasonable rounding.)

$$d) \text{proj}_{\vec{b}} \vec{a} = \left(\frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} \right) \vec{b}$$

$$= \left(\frac{6\lambda - 6}{40} \right) \begin{bmatrix} 6 \\ -2 \end{bmatrix} \quad \checkmark$$

$$\text{length} = \left| \frac{6\lambda - 6}{40} \right| \sqrt{40} \quad \checkmark$$

$$\text{length} = \frac{|6\lambda - 6|}{\sqrt{40}} = 3\sqrt{10}$$

$$|6\lambda - 6| = 60$$

$$6\lambda - 6 = 60$$

$$\lambda - 1 = 10$$

$$\lambda = 11$$

$$\text{or } 6\lambda - 6 = -60$$

$$\lambda - 1 = -10$$

$$\lambda = -9 \quad \checkmark$$

(Both neg.
for last
mark).

Alternatively:

$$\vec{a} = \vec{a}_i + 3\vec{j} \quad \vec{b} = 6\vec{i} - 2\vec{j}$$

$$|\text{proj}_{\vec{b}} \vec{a}| = \left| \frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} \right| \times |\vec{b}|$$

$$\vec{a} \cdot \vec{b} = \begin{bmatrix} \lambda \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ -2 \end{bmatrix} = 6\lambda - 6$$

$$\vec{b} \cdot \vec{b} = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ -2 \end{bmatrix} = 36 + 4 = 40$$

$$\text{proj}_{\vec{b}} \vec{a} = \frac{6\lambda - 6}{40} \times \begin{bmatrix} 6 \\ -2 \end{bmatrix} \quad \checkmark \quad \frac{3\lambda - 3}{20} (6\vec{i} - 2\vec{j})$$

$$= \frac{3\lambda - 3}{10} (3\vec{i} - \vec{j})$$

$$|\text{proj}_{\vec{b}} \vec{a}| = \left| \frac{3\lambda - 3}{10} (3\vec{i} - \vec{j}) \right|$$

$$= \left| \frac{3\lambda - 3}{20} \times \sqrt{40} \right|$$

$$= \left| \frac{(3\lambda - 3) \times 2\sqrt{10}}{20} \right|$$

$$= \left| \frac{(3\lambda - 3)\sqrt{10}}{10} \right|$$

$$\left| \frac{(3\lambda - 3)\sqrt{10}}{10} \right| = 3\sqrt{10} \quad \checkmark$$

$$\frac{(3\lambda - 3)\sqrt{10}}{10} = 3\sqrt{10}$$

$$\frac{3\lambda - 3}{10} = 3$$

$$\frac{\lambda - 1}{10} = 1$$

$$\lambda - 1 = 10$$

$$\lambda = 11$$

$$\frac{(3\lambda - 3)\sqrt{10}}{10} = -3\sqrt{10}$$

$$\frac{\lambda - 1}{10} = -1$$

$$\lambda - 1 = -10$$

$$\lambda = -9 \quad (\checkmark)$$

(Both
neg.)

$$13). a) \cos^2 nx = \frac{1}{2}(1 + \cos 2nx) \quad \frac{\pi}{4} < \pi/3$$

$$\begin{aligned}
 V &= \int_{\pi/4}^{\pi/3} \pi y^2 dx = \pi \int_{\pi/4}^{\pi/3} \cos^2 x dx \\
 &= \pi \int_{\pi/4}^{\pi/3} \frac{1}{2}(1 + \cos 2x) dx \quad \checkmark \\
 &= \frac{\pi}{2} \int_{\pi/4}^{\pi/3} 1 + \cos 2x dx \\
 &= \frac{\pi}{2} \left[x + \frac{1}{2} \sin 2x \right]_{\pi/4}^{\pi/3} \quad \checkmark \\
 &= \frac{\pi}{2} \left[\left(\frac{\pi}{3} + \frac{1}{2} \sin(2\pi/3) \right) - \left(\frac{\pi}{4} + \frac{1}{2} \sin(2\pi/4) \right) \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{3} + \frac{1}{2} \sin(\pi/3) - \frac{\pi}{4} - \frac{1}{2} \sin(\pi/2) \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{12} + \frac{1}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times 1 \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{12} + \frac{\sqrt{3}}{4} - \frac{1}{2} \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{12} + \frac{3\sqrt{3}}{12} - \frac{6}{12} \right] \\
 &= \frac{\pi}{24} (\pi + 3\sqrt{3} - 6) \quad u^3 \quad \checkmark
 \end{aligned}$$

b) i) Show $\frac{dV}{dt} = -k_2 V^{2/3}$ where $k_2 = k_1 (36\pi)^{1/3}$

Given: $\frac{dV}{dt} = -k_1 A$

$$A = 4\pi r^2 \quad V = \frac{4\pi r^3}{3}$$

$$\frac{3V}{4\pi} = r^3$$

$$r = \left(\frac{3V}{4\pi} \right)^{1/3}$$

$$\therefore \frac{dV}{dt} = -k_1 \times 4\pi \left(\frac{3V}{4\pi} \right)^{2/3} \quad \checkmark$$

$$\frac{dV}{dt} = -k_1 \times \frac{4\pi}{(4\pi)^{2/3}} \times 3^{2/3} \times V^{2/3}$$

$$\frac{dV}{dt} = -k_1 \times (4\pi)^{1/3} \times 9^{1/3} \times V^{2/3} \quad \checkmark$$

$$\frac{dV}{dt} = -k_1 (36\pi)^{1/3} \times V^{2/3}$$

$$\therefore \frac{dV}{dt} = -k_2 V^{2/3} \quad \text{where } k_2 = k_1 (36\pi)^{1/3}$$

ii)

$$\frac{dV}{dt} = -k_2 V^{2/3}$$
$$\int V^{-2/3} dV = \int -k_2 dt$$

$$3V^{1/3} = -k_2 t + C$$

at $t=0 \quad V=V_0$

$$3V_0^{1/3} = C$$

$$\therefore 3V^{1/3} = -k_2 t + 3V_0^{1/3}$$
$$3V^{1/3} = 3V_0^{1/3} - k_2 t$$
$$V^{1/3} = V_0^{1/3} - \frac{1}{3}k_2 t$$
$$V = \left(V_0^{1/3} - \frac{1}{3}k_2 t\right)^3$$

iii)

when $t=6 \quad V = \frac{V_0}{8}$

$$\frac{V_0}{8} = \left(V_0^{1/3} - \frac{1}{3} \times k_2 \times 6\right)^3$$

$$\frac{V_0^{1/3}}{2} = V_0^{1/3} - 2k_2$$

$$2k_2 = \frac{V_0^{1/3}}{2}$$

$$k_2 = \frac{V_0^{1/3}}{4}$$

iv)

99% dissolved

$$\therefore V = \frac{V_0}{100}$$

$$V = \left(V_0^{1/3} - \frac{1}{3} k_2 t \right)^3$$

$$k_2 = \frac{V_0^{1/3}}{4}$$

$$\frac{V_0}{100} = \left(V_0^{1/3} - \frac{1}{3} \times \frac{V_0^{1/3}}{4} \times t \right)^3$$

$$\frac{V_0}{100} = \left(V_0^{1/3} \left(1 - \frac{t}{12} \right) \right)^3$$

$$\frac{V_0}{100} = V_0 \left(1 - \frac{t}{12} \right)^3$$

$$\frac{1}{100} = \left(1 - \frac{t}{12} \right)^3$$

$$\frac{1}{\sqrt[3]{100}} = 1 - \frac{t}{12}$$

$$\frac{t}{12} = 1 - \frac{1}{\sqrt[3]{100}}$$

$$t = 12 \left(1 - \frac{1}{\sqrt[3]{100}} \right)$$

$$t \approx 9.414678 \dots \text{ hrs}$$

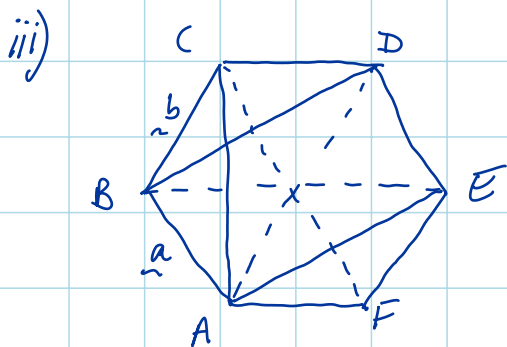
$$t = 9 \text{ hours and } 25 \text{ minutes.} \quad \checkmark$$

(Do not penalise rounding despite what the question says.)

$$c) \ i) \quad |\vec{AP}| : |\vec{PC}| \\ \lambda : 1 - \lambda \quad \checkmark$$

$$ii) \quad \vec{AC} = \underline{a} + \underline{b}$$

$$\vec{AP} = \lambda \times \vec{AC} \\ = \lambda(\underline{a} + \underline{b}) \quad \checkmark$$



$$\vec{BD} = \vec{AE} \\ = 2\underline{b} - \underline{a} \quad \checkmark$$

$$iv) \quad \vec{BP} = (1 - \lambda) \vec{BD} \\ = (1 - \lambda)(2\underline{b} - \underline{a}) \quad \checkmark$$

$$\therefore \vec{AP} = \vec{AB} + \vec{BP} \\ = \underline{a} + (1 - \lambda)(2\underline{b} - \underline{a}) \\ = \underline{a} + 2\underline{b} - \underline{a} - 2\lambda\underline{b} + \lambda\underline{a} \\ = \lambda\underline{a} + (2 - 2\lambda)\underline{b}$$

but from ii) $\vec{AP} = \lambda\underline{a} + \lambda\underline{b}$

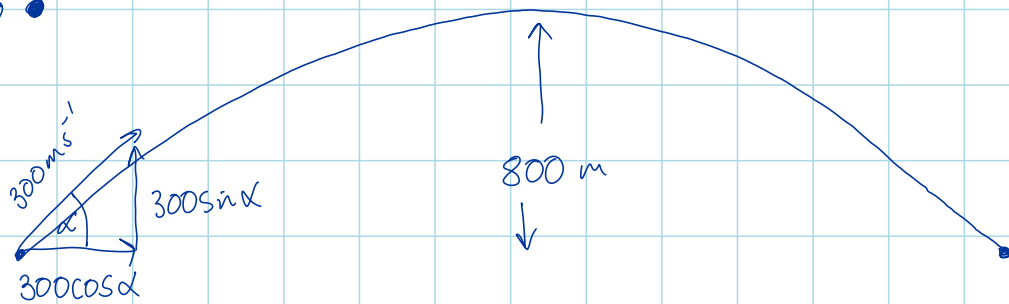
Comparing coefficients of $\underline{b}$: $\lambda = 2 - 2\lambda$

$$3\lambda = 2$$

$$\lambda = \frac{2}{3} \quad \checkmark$$

14) a) i)

$108 \text{ km/h} = 30 \text{ m/s}$



Let angle to horizontal be α .

To hit, horizontal displacement of both drone and bullet must be equal.

Drone: $\dot{x} = 30$
 $x = 30t$

Bullet: $\dot{x} = 0$
 $\dot{x} = 300 \cos \alpha$
 $x = 300t \cos \alpha$

$\left\{ \begin{array}{l} \text{both} \\ \text{expressions} \\ \text{for } x. \end{array} \right.$

$\therefore 300t \cos \alpha = 30t$ to hit

$\cos \alpha = \frac{1}{10}$

$\alpha = \cos^{-1}(\frac{1}{10})$

$\approx 84^\circ$ (to nearest degree)

ii)

Bullet

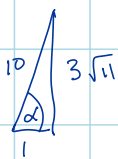
Vertically: $\ddot{y} = -10$

$\dot{y} = -10t + 300 \sin \alpha$

$y = -5t^2 + 300t \sin \alpha$

$\cos \alpha = \frac{1}{10}$

$\sin \alpha = \frac{3\sqrt{11}}{10}$



$y = -5t^2 + 300t \sin \alpha$

$y = -5t^2 + 300t \left(\frac{3\sqrt{11}}{10} \right)$

$y = -5t^2 + 90\sqrt{11}t$

Max ht when $\dot{y} = 0$ $-10t + 300 \sin \alpha = 0$

$t = 30 \sin \alpha$
 $= 30 \frac{3\sqrt{11}}{10}$

[or

$$= 9\sqrt{11} \text{ sels}$$

Find when $y=0$, use symmetry to find when $y=y_{\max}$
 let $y=0$

$$0 = -5t^2 + 90\sqrt{11}t$$

$$0 = t^2 - 18\sqrt{11}t$$

$$0 = t(t - 18\sqrt{11})$$

$$t=0$$

and

$$t=18\sqrt{11}$$

gives $y=0$ $y_{\max}$

when

$$t = \frac{0+18\sqrt{11}}{2} = 9\sqrt{11}$$

seconds

$$y_{\max} = -5(9\sqrt{11})^2 + 90\sqrt{11}(9\sqrt{11})$$

$$= -5 \times 81 \times 11 + 810 \times 11$$

$$= 4455 \text{ m}$$

$$= 4.455 \text{ km}$$

b) i)

Tadpoles:

$$0 < P < K$$

$$P < K$$

$$\frac{P}{K} < 1$$

$$0 < 1 - \frac{P}{K}$$

$$1 - \frac{P}{K} > 0$$

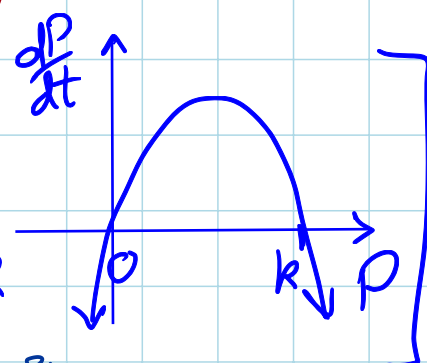
$$\therefore rP(1 - \frac{P}{K}) > 0$$

$$\therefore \frac{dP}{dt} > 0$$

$$r, P > 0$$

$$\text{or } \frac{dP}{dt} > 0$$

$$\text{for } 0 < P < K$$



logical argument.

$$\text{ii) } \frac{dP}{dt} = rP(1 - \frac{P}{K})$$

$$K \cdot \frac{dP}{dt} = rP(1 - \frac{P}{K})K$$

$$K \cdot \frac{dP}{dt} = rP(K - P)$$

$$\int \frac{K}{P(K-P)} dP = \int r dt$$

$$\text{Since } \frac{K}{P(K-P)} = \frac{1}{P} + \frac{1}{K-P}$$

$$\int \frac{1}{P} + \frac{1}{K-P} dP = rt + C_1$$

$$\ln |P| - \ln |K-P| = rt + C_1$$

$$\ln \left| \frac{P}{K-P} \right| = rt + C_1$$

$$-\ln \left| \frac{K-P}{P} \right| = rt + C_2$$

$$\ln \left| \frac{K}{P} - 1 \right| = -rt + C_2$$

$$\frac{K}{P} - 1 = e^{-rt + C_2}$$

$$\frac{K}{P} = 1 + e^{-rt + C_2}$$

$$\text{Let } A = e^{C_2}$$

$$\frac{K}{P} = 1 + Ae^{-rt}$$

$$P = \frac{K}{1 + Ae^{-rt}}$$

$$\text{at } t=0, P=P_0$$

$$P_0 = \frac{K}{1+A}$$

$$1+A = \frac{K}{P_0}$$

$$A = \frac{k}{P_0} - 1$$

$$A = \frac{k}{P_0} - \frac{P_0}{P_0}$$

$$A = \frac{k - P_0}{P_0}$$

$$\therefore P = \frac{k}{1 + \left(\frac{k - P_0}{P_0}\right)e^{-rt}}$$

$$P = \frac{k P_0}{P_0 + (k - P_0)e^{-rt}} \quad \checkmark$$

as required.

Requires clear,
full, logical
progression
for 3 marks.

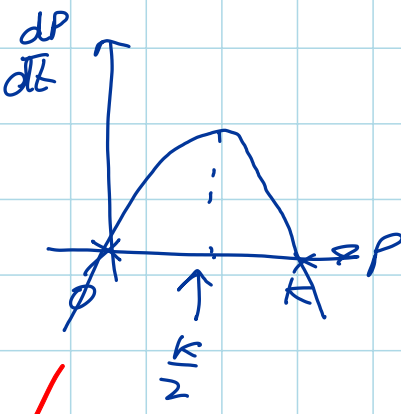
$$\begin{aligned} \text{iii) } \lim_{t \rightarrow \infty} P &= \lim_{t \rightarrow \infty} \frac{k P_0}{P_0 + (k - P_0)e^{-rt}} \\ &= \frac{k P_0}{P_0} \\ &= k \end{aligned}$$

$\checkmark$ Needs
justification

iv) $\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$

$P=0$ or $1 - \frac{P}{K} = 0$
 $P=K$

maximum at $P = \frac{K}{2}$ ✓



$$P = \frac{KP_0}{P_0 + (K - P_0)e^{-rt}}$$

When $P = \frac{K}{2}$

$$\frac{K}{2} = \frac{KP_0}{P_0 + (K - P_0)e^{-rt}}$$

$$\frac{1}{2} = \frac{P_0}{P_0 + (K - P_0)e^{-rt}}$$

$$P_0 + (K - P_0)e^{-rt} = 2P_0$$

$$(K - P_0)e^{-rt} = P_0$$

$$e^{-rt} = \frac{P_0}{K - P_0}$$

$$-rt = \ln \left| \frac{P_0}{K - P_0} \right|$$

$$t = -\frac{1}{r} \ln \left| \frac{P_0}{K - P_0} \right|$$

$$t = \frac{1}{r} \ln \left| \frac{K - P_0}{P_0} \right|$$
 ✓

c) Prove $\sin x + \sin(x+y) + \sin(x+2y) + \dots + \sin(x+(n-1)y)$

$$= \frac{\sin(x + \frac{1}{2}(n-1)y) \sin(\frac{1}{2}ny)}{\sin(\frac{1}{2}y)}$$

 for $n > 1$

A. let $n = 1$

$$\text{LHS} = \sin(x)$$

$$\text{RHS} = \frac{\sin(x + \frac{1}{2} \times 0 \times y) \sin(\frac{1}{2}y)}{\sin(\frac{1}{2}y)}$$

$$= \sin(x)$$

$$= \text{LHS.}$$

The statement is true for $n = 1$.

B. Assume true for $n = k$. Assume

$$\sin x + \sin(x+y) + \sin(x+2y) + \dots + \sin(x+(k-1)y)$$

$$= \frac{\sin(x + \frac{k-1}{2}y) \sin(\frac{1}{2}ky)}{\sin(\frac{1}{2}y)}$$

Now prove for $n = k+1$, prove

$$\sin(x) + \sin(x+y) + \sin(x+2y) + \dots + \sin(x+(k-1)y) + \sin(x+ky)$$

$$= \frac{\sin(x + \frac{1}{2}ky) \sin(\frac{1}{2}(k+1)y)}{\sin(\frac{1}{2}y)}$$

$$\text{LHS} = \sin(x) + \sin(x+y) + \sin(x+2y) + \dots + \sin(x+(k-1)y) + \sin(x+ky)$$

$$= \frac{\sin(x + \frac{1}{2}(k-1)y) \sin(\frac{1}{2}ky)}{\sin(\frac{1}{2}y)} + \sin(x+ky)$$

✓
Set up
Correct

* By the induction hypothesis.

$$= \frac{\sin(x + \frac{1}{2}(k-1)y) \sin(\frac{1}{2}ky)}{\sin(\frac{1}{2}y)} + \frac{\sin(x+ky) \sin(\frac{1}{2}y)}{\sin(\frac{1}{2}y)}$$

$$= \frac{\sin(x + \frac{1}{2}(k-1)y) \sin(\frac{1}{2}ky) + \sin(x+ky) \sin(\frac{1}{2}y)}{\sin(\frac{1}{2}y)}$$

$$\sin(A)\sin(B) = \frac{1}{2}(\cos(A-B) - \cos(A+B))$$

$$= \frac{1}{2} \left[\cos\left(x + \frac{k-1}{2}y - \frac{ky}{2}\right) - \cos\left(x + \frac{k-1}{2}y + \frac{ky}{2}\right) \right]$$

$$+ \frac{1}{2} \left[\cos\left(x + ky - \frac{y}{2}\right) - \cos\left(x + ky + \frac{y}{2}\right) \right]$$

$$\frac{\sin(\frac{y}{2})}{\sin(\frac{y}{2})}$$

$$= \frac{\frac{1}{2} \cos\left(x - \frac{y}{2}\right) - \frac{1}{2} \cos\left(x + y\left(\frac{2k-1}{2}\right)\right) + \frac{1}{2} \cos\left(x + y\left(\frac{2k-1}{2}\right)\right) - \frac{1}{2} \cos\left(x + y\left(\frac{2k+1}{2}\right)\right)}{\sin(\frac{y}{2})}$$

$$= \frac{\frac{1}{2} \cos\left(x - \frac{y}{2}\right) - \frac{1}{2} \cos\left(x + (k+\frac{1}{2})y\right)}{\sin \frac{y}{2}}$$

$$= \frac{\frac{1}{2} \left[\cos\left(x + \frac{ky}{2} - \left(\frac{ky}{2} + \frac{y}{2}\right)\right) - \cos\left(x + \frac{ky}{2} + \left(\frac{ky}{2} + \frac{y}{2}\right)\right) \right]}{\sin \frac{y}{2}}$$

$$= \frac{\sin\left(x + \frac{ky}{2}\right) \sin\left(\frac{ky}{2} + \frac{y}{2}\right)}{\sin \frac{y}{2}}$$

$$= \text{RHS}$$

Using product to sums
where $A = \left(x + \frac{ky}{2}\right)$
 $B = \left(\frac{ky}{2} + \frac{y}{2}\right)$

By Principle of Mathematical Induction result true for all positive integer n .

✓ Use of Product to Sums

✓ use of sum to product

C. By the process of mathematical induction,
the statement is true for $n=1$,
 $n=k$ and $n=k+1$, therefore
is true for all $n \geq 1$.